

NEP89: Universal neuroevolution potential for inorganic and organic materials across 89 elements

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While machine-learned interatomic potentials offer near-quantum-mechanical accuracy for atomistic simulations, many are material-specific or computationally intensive, limiting their broader use. Here, we introduce NEP89, a foundation model based on neuroevolution potential architecture, delivering near-empirical-potential speed and high accuracy across 89 elements. A compact yet comprehensive training dataset covering inorganic and organic materials was curated through descriptor-space subsampling and iterative refinement across multiple datasets. NEP89 achieves competitive accuracy compared to representative foundation models while being three to four orders of magnitude more computationally efficient, enabling previously impractical large-scale atomistic simulations of inorganic and organic systems. In addition to its out-of-the-box applicability to diverse scenarios, including million-atom-scale compression of compositionally complex alloys, ion diffusion in solid-state electrolytes and water, rocksalt dissolution, methane combustion, and protein-ligand dynamics, NEP89 also supports fine-tuning for rapid adaptation to user-specific applications, such as mechanical, thermal, structural, and spectral properties of two-dimensional materials, metallic glasses, and organic crystals.

I. INTRODUCTION

Large-scale atomistic molecular dynamics (MD) simulations are invaluable tools for elucidating the intricate properties of complex materials. However, the fidelity of these simulations critically relies on the accuracy of the underlying interatomic potentials (or force fields). *Ab initio* molecular dynamics (AIMD) simulations offer the desired level of fidelity, but their computational cost is prohibitive in large-scale MD simulations of millions of atoms and more.

In recent years, artificial intelligence has been successfully applied to construct accurate yet efficient machine-

learned interatomic potential models [1, 2], substantially advancing the field of atomistic simulations [3–7]. Early models were developed for specific materials and even specific properties of a given material. More recent developments have revealed a trend toward building more general-purpose potentials, from a single element [8] to a set of metals [9], to tens of elements [10, 11], and to almost the entire periodic table [12–17]. The last category of potential models, although encompassing many species, remains incomplete, as they were mostly constructed based on inorganic materials. Moreover, their high computational cost severely limits their practical applicability, particularly for large-scale or long-timescale simulations.

Here, to overcome these limitations, we introduce NEP89, a foundation model that enables high-fidelity simulations of both inorganic and organic materials across 89 elements, while achieving a computational speed several orders of magnitude faster than previous universal interatomic potential models. Our model is based on the state-of-the-art neuroevolution potential (NEP) approach [9, 18], which combines high accuracy with efficiency. This approach has recently been demonstrated to achieve a general-purpose interatomic potential for 16 metals and their alloys [9].

One major obstacle is the lack of a readily usable training dataset that comprehensively incorporates both inorganic and organic materials. The lack of such a dataset is presumably the reason why previous works have only trained universal potential models either for inorganic [12–17] or organic materials [19–22], or both, but with multiple training tasks instead of a single unified one [11]. A crucial contribution in the present work is the curation of existing databases of separate sets of materials to ensure they have consistent target data for training a unified interatomic potential for both inorganic and organic materials. Starting from a sub-sampled dataset of OMat24 [23], we iteratively train NEP89 and build the corresponding training dataset, incorporating structures from many other public datasets [9, 13, 19–21, 24–26]. Moreover, we add the D3 dispersion correction [27, 28] to some datasets such that all of them have proper dispersion interactions. Furthermore, the relative energies between different datasets are also optimized during the training process, which is a prerequisite for constructing a unified single-task model. The NEP89 model achieves competitive accuracy in predicting both static and dynamic properties. It also enables large-scale atomistic simulations of both inorganic and organic systems out of the box, while supporting rapid and cost-effective fine-tuning across a broad range of applications.

II. RESULTS

The architecture of the NEP89 model and its finetuning. The NEP89 model is built on the NEP approach [9, 18]. We begin by briefly introducing the NEP architecture and then discuss the extensions that enable the fine-tuning of NEP89. The NEP approach is a many-body neural network potential that uses an atom-decomposed total energy and atom-centered descriptors constructed with Chebyshev and Legendre polynomials, along with an advanced training algorithm that incorporates regularization and employs the separable natural evolution strategy [29]. The latter evolves a set of mean and variance values for the trainable parameters that also form the basis for fine-tuning. The descriptors are defined within a specified cutoff radius to ensure that the computational cost scales linearly with the number of atoms. The input layer neurons of the neural network correspond to the descriptors, while the output

layer neuron represents the site energy U_i of the central atom i . Force and virial stress are calculated via analytical derivatives of the energy (see [Supplementary Section 1](#) for details). Atomic species are encoded in the expansion coefficients of a set of radial functions. For each pair of species, there is an independent set of N_{ec} expansion coefficients optimized during training. Additionally, each species has its own independent set of N_{nn} trainable weight and bias parameters within the neural network. Therefore, for a system with N_{spe} species, there are $N_{\text{spe}}^2 N_{\text{ec}} + N_{\text{spe}} N_{\text{nn}}$ trainable parameters (refer to [Supplementary Section 2](#) for the hyperparameters used when training NEP89).

Thanks to the combinatorial architecture of NEP, the trained NEP89 model can be conveniently fine-tuned into smaller models tailored to subsets of species using a new training dataset. During the fine-tuning process, it is crucial to maintain the descriptor normalization. To this end, for the targeted species, the mean and variance of their trainable parameters, used by the separable natural evolution strategy, are extracted from NEP89 and reused during continued training. Fine-tuning typically requires only a small fraction of the training steps compared to training from scratch to achieve optimal results. Additionally, to prevent the model from undesirably losing memory, the variance values of the descriptor parameters can be set to zero, ensuring that only the neural network parameters are updated during fine-tuning.

Iterative training of NEP89. To train a universal potential model across 89 species for both inorganic and organic materials, the first task is to construct a training set that is diverse, reliable, and consistent. So far, no single available dataset meets these three conditions simultaneously. Fortunately, there are already many individual datasets, and we aim to combine them to form a consistent dataset with enhanced diversity and reliability. A notable one is the OMat24 dataset [23] containing over 110 million structures for inorganic bulk materials. To supplement the OMat24 dataset, we also include structures from other important datasets, including the MPtrj dataset for relaxation trajectories of inorganic bulk materials across 89 elements [13], the SPICE dataset for drug-like small molecules, peptides, and solvated amino acids [20], the ANI-1xn dataset for reactive organics [21], the SSE-ABACUS and SSE-VASP datasets for solid-state electrolytes [24], the solvated protein fragments (Protein for short) dataset [19], the UNEP-v1 dataset for 16 metals and their alloys [9], the CH dataset for general and reactive CH systems [25], the water dataset with liquid, gas, and vapor-liquid phases [26], and a reactive CHONPS dataset we created. More details regarding these datasets are provided in [Supplementary Table 1](#). The successful training of NEP89 was not accomplished by providing all the aforementioned datasets to the optimizer simultaneously, but rather through an iterative training strategy, as described below.

Initially, we note that these datasets are inconsistent

in their incorporation of dispersion interactions. While some datasets (SPICE, Protein, ANI-1xnr) include the D3 dispersion [27, 28], we added this dispersion interaction to the remaining datasets for uniformity. The water dataset is unique, as it was labeled using the MB-pol method [30], delivering accuracy close to CCSD(T). Although not all datasets are calculated at the same level of quantum-mechanical theory, the differences in force and stress are minor and can be disregarded in training the NEP89 model. On the other hand, these datasets have substantially different energy references for a given species, which hinders the development of a unified model. To address this challenge, we initially focused exclusively on the energies in the OMAT24 dataset, disregarding reference energies from all other datasets.

Another challenge in training a unified model based on extensive public datasets is managing the vast number of training structures. A good training dataset does not necessarily require a large number of structures but rather a diverse and well-balanced selection. To address this, we first created a randomly sub-sampled dataset from the OMAT24 database to train an initial model. Using this model, we then selected a subset of the OMAT24 database with uniformly distributed descriptor values (see [Supplementary Section 3](#) for details) to train an improved model. We then performed model training and data curation for more than ten iterations. During each iteration, the current model predicted unseen structures across all datasets. Structures with high force and stress errors were identified, and a subset with evenly distributed descriptor values was sampled and added to the training dataset. This refined dataset was then used to train the next model in the sequence.

An essential aspect of dataset refinement is identifying inaccuracies in the reference data. Several datasets, such as OMAT24 and MPtrj, contain erroneous entries or unconverged density functional theory (DFT) calculations. To mitigate the impact of noisy reference data, prior studies have employed the Huber loss function [12–14, 16, 17]. Our approach involves iteratively removing outliers from the training dataset, thereby gradually reducing their influence on the resulting models. In [Supplementary Figure 1](#), we show results for 113 typical structures in the MPtrj dataset identified as outliers during our training, where the original DFT reference values were found to be inaccurate, while predictions by NEP89 closely aligned with our recalculated reference values.

Through this iterative training and refinement process, we developed a diverse and balanced training dataset, which is visualized in the reduced descriptor space spanned by the two principal components in [Figure 1a](#). However, the model obtained in this process was exclusively trained on the OMAT24 dataset in terms of energy, introducing a possible bias toward this dataset. To ensure balanced training across different datasets, we accordingly adjusted the reference energies for the other datasets. These adjustments minimized differences between the model predictions and the shifted reference

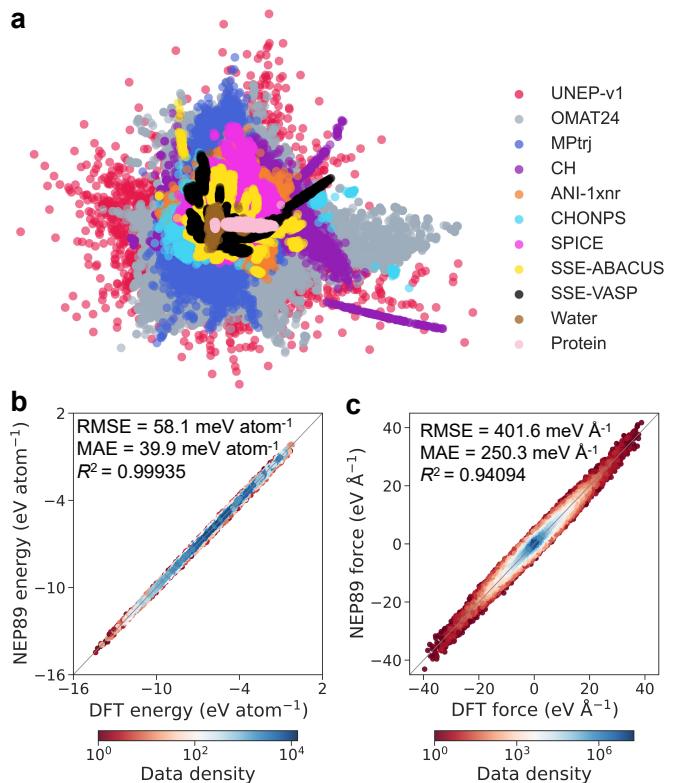


FIG. 1. Dataset composition for NEP89 and overall training accuracy. **a**, Distribution of the NEP89 datasets in the reduced descriptor space spanned by the first two principal components. The individual datasets were subsampled from the UNEP-v1 [9], OMAT24 [23], MPtrj [13], CH [25], ANI-1xnr [21], SPICE [20], SSE-ABACUS [24], SSE-VASP [24], water [26], and Protein [19] datasets, or specifically prepared in this work (CHONPS). **b,c**, Parity plots for energy and force comparing NEP89 predictions with reference values (DFT or high-level quantum chemistry). The color intensity visualizes the distribution and density of the NEP89 datasets.

energies (see [Supplementary Section 4](#) for details). We excluded the reference energies for the MPtrj dataset from training, since they were incompatible with those in the OMAT24 dataset due to differing treatments of the Hubbard U correction. We then performed the energy adjustment for a few iterations and trained the final model, which we refer to as the NEP89 model. The parity plot for the energies ([Figure 1b](#)) indicates that all the considered reference energies are indeed quite consistent after our adjustment. The parity plot for the forces ([Figure 1c](#)) also indicates a well-trained model for the combined dataset. ([Supplementary Figure 2](#) shows error metrics for energy, force, and stress for the individual datasets.)

Benchmarks on static properties. [Figure 2](#) presents a number of benchmarks on static properties predicted by NEP89 and various foundation models, including MACE-MP-0 (medium version) [16], CHGNet [13] (ver-

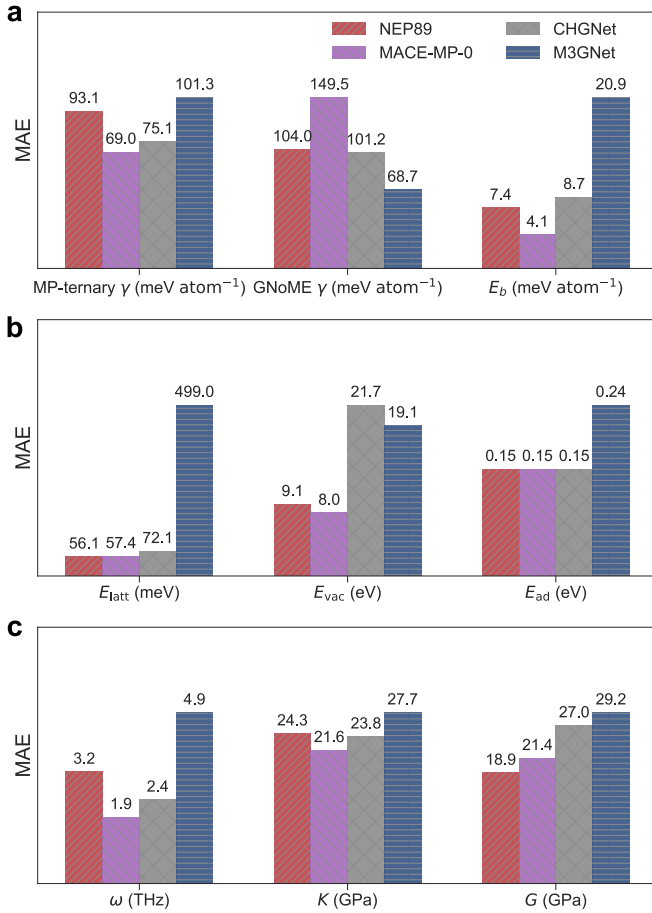


FIG. 2. **Various benchmarks on static properties for validating the foundation models.** Mean absolute errors (MAEs) of several foundation models, NEP89 (this work), MACE-MP-0 [16], CHGNet [13], and M3GNet [12], with respect to DFT reference data for nine evaluated properties. **a**, Formation energies (γ) for structures sampled from the Materials Project (MP-ternary) [31] and the GNoME dataset [14], and binding energies (E_b) of the S66 dimer set [32] with DFT (PBE+D3) references [16]. **b**, Lattice energies (E_{latt}) for the DMC-ICE13 dataset [33] with DFT references [16], formation energies (E_{vac}) of iron vacancy clusters of different sizes, and adhesion energy (E_{ad}) of hydrogen atoms in iron nanopores with DFT references (PBE+D3). **c**, Highest phonon band frequencies (ω) of 97 materials with DFT (PBE) references from the PhononDB database [34], and bulk (K) and shear (G) moduli with DFT references from Materials Project [31] covering more than 11,000 materials.

sion 0.3.0), and M3GNet [12] (version 2021.2.8-DIRECT-PES); see ‘Benchmark calculations for static properties’ section in Methods, [Supplementary Sections 5–10](#), and [Supplementary Figures 3–8](#) for calculation details. Overall, NEP89 demonstrates comparable accuracy to other foundation models, based on mean absolute errors across nine evaluated properties relative to DFT data. For the formation energies (γ) of the ternary structures sampled from the Materials Project (MP-ternary) [31], MACE-MP-0 and CHGNet are slightly more accurate, with

NEP89 ranking third, despite not being trained on the MPtrj energies. For the 2-component to 5-component structures from the GNoME dataset [14], on which none of the models were trained, MACE-MP-0 shows the worst accuracy, while NEP89 performs very close to CHGNet as the second-best model. For the binding energies (E_b) of the S66 dimer set [32], NEP89 ranks as the second-best model, even though it does not require a separate D3 correction as for the other models. NEP89 demonstrates superior accuracy for lattice energies (E_{latt}) on the DMC-ICE13 dataset [16, 33], second-best accuracy for formation energies (E_{vac}) of iron vacancy clusters of different sizes, and comparable accuracy for adhesion energy (E_{ad}) of hydrogen atoms in iron nanopores (see [Supplementary Section 8](#) for details on our iron vacancy cluster and nanopore datasets). For the highest phonon band frequencies (ω) for 97 materials from the PhononDB database [34], NEP89 achieves the third-best performance. For more than 11 000 materials from Materials Project [31], NEP89 demonstrates comparable accuracy in predicting the bulk modulus (K) and the best accuracy in predicting the shear modulus (G).

Evaluation of computational performance. Since the primary motivation for developing machine-learned interatomic potentials is to extend the spatiotemporal scales of ab initio calculations, computational efficiency, in terms of both memory usage and computational speed, is a crucial factor. As shown in [Figure 3a–b](#), NEP89 consistently outperforms other foundation models in terms of speed and memory usage across a wide range of system sizes, using a 20-element high-entropy alloy as a representative example (see ‘Computational performance evaluation’ section in Methods for details). On the 24-GB RTX 4090 and 80-GB H800 GPUs considered here, NEP89 can simulate up to 8×10^6 atoms at a speed of 5×10^6 atom-steps per second and 1.5×10^7 atoms at a speed of 7×10^6 atom-steps per second, respectively. Overall, NEP89 extends the accessible spatiotemporal scale of atomistic simulations by at least three orders of magnitude compared to other foundation models, while maintaining competitive accuracy for both static properties (as shown above) and dynamic properties (as we will discuss next). In addition, NEP89 exhibits efficient multi-GPU scalability ([Supplementary Figure 9](#)). Benchmarks on water systems ([Supplementary Figure 10](#)) shows that NEP89 attains simulation speeds approaching those of empirical water potentials, substantially narrowing the performance gap between empirical potentials and foundation models.

Benchmarks of dynamical properties. To further evaluate the performance of NEP89 in atomistic MD simulations, we compare dynamical properties of three typical systems with different structural phases, including amorphous carbon, liquid water, and three solid-state electrolytes, obtained using NEP89 and other foundation models (see ‘MD simulation of amorphous carbon’, ‘MD

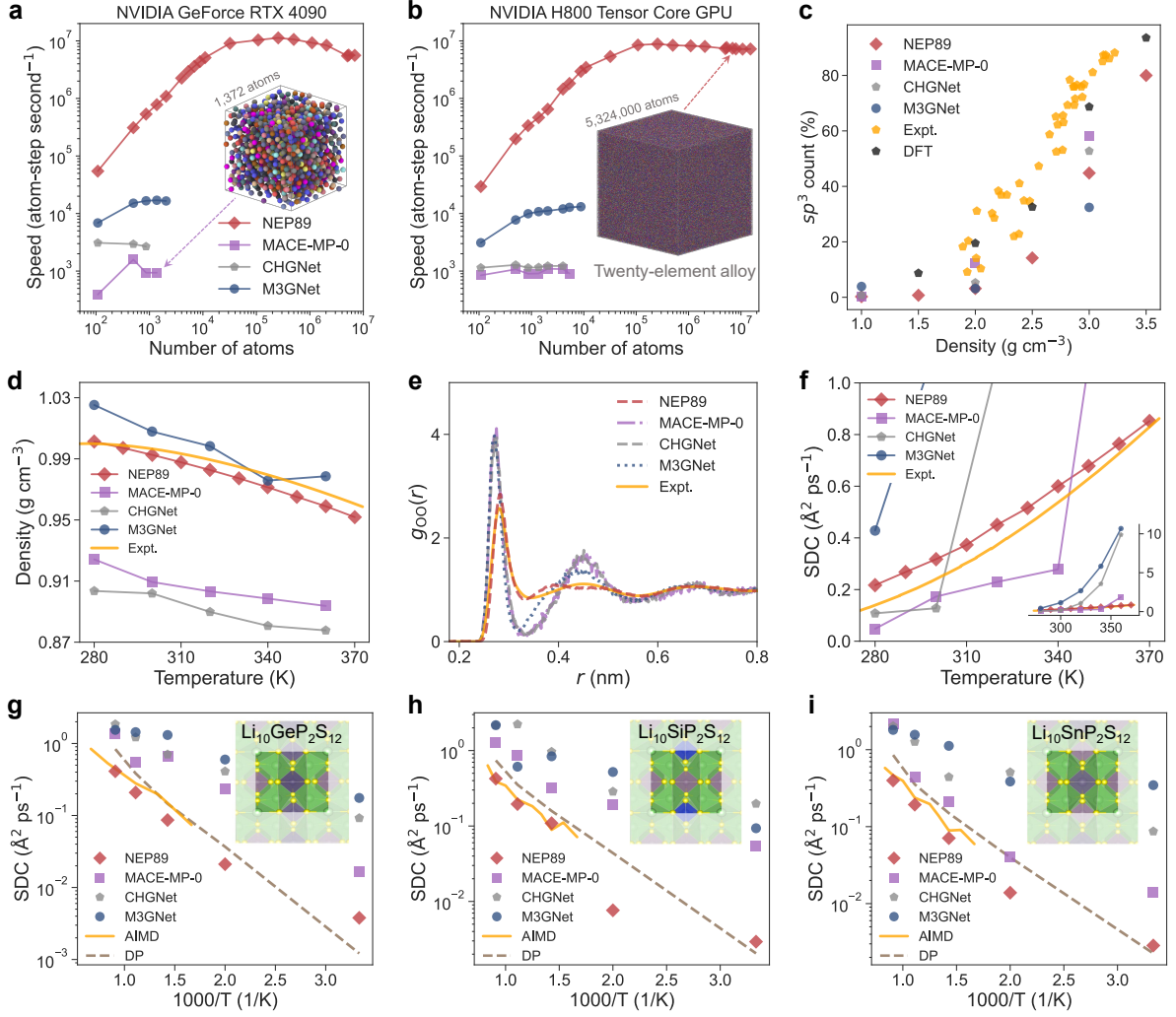


FIG. 3. Computational performance and benchmarks of dynamical properties. **a,b**, Benchmark results for computational efficiency of NEP89 and various foundation models in twenty-element alloys. Insets show representative atomic snapshots of the alloy model used in the MD simulations. NEP89 achieves about a three-order-of-magnitude improvement in computational efficiency compared to the other foundation models. **c**, Comparison of experimental and predicted bonding statistics of amorphous carbon. **d–f**, Experimental and predicted density (**d**), radial distribution function (**e**), and self-diffusion coefficient (SDC) (**f**) of liquid water simulated using NEP89 and other foundation models. **g–i**, Experimental and calculated lithium-ion SDC in (**g**) $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, (**h**) $\text{Li}_{10}\text{SiP}_2\text{S}_{12}$, and (**i**) $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$ solid-state electrolytes using NEP89, three other foundation models, a specialized deep potential (DP) model, and AIMD. Overall, predictions from NEP89 show good agreement with experimental data or AIMD results.

simulation of water’, and ‘Lithium-ion diffusion in solid-state electrolytes’ sections in Methods for computational details). Figure 3c shows the bonding statistics of amorphous carbon, comparing experimental data [35] with predictions from NEP89 and other foundation models. NEP89 achieves comparable accuracy to other foundation models and aligns reasonably well with experimental results. Figure 3d–f shows a comparative analysis of the density, radial distribution function, and self-diffusion coefficient of liquid water as obtained from experiments [36], NEP89, and other foundation models. The results show that the NEP89 predictions closely match experimental data and that NEP89 outperforms the other founda-

tion models, particularly in reproducing the density (Figure 3d) and the oxygen-oxygen radial distribution function (Figure 3e). As further illustrated in Supplementary Figure 11, NEP89 produces balanced water density predictions that, at certain temperatures, align more closely with experimental data than MB-pol, one of the most accurate models for water [36]. While this behavior is not systematic across all conditions, it suggests the potential role of partial error compensation across heterogeneous data sources enabled by the reference-energy adjustment strategy (see Supplementary Sections 4 and 11). In Figure 3e, a minor discrepancy near $r = 0.4$ nm is likely attributable to slight underfitting

during the global optimization of NEP89; this discrepancy can be further reduced by fine-tuning on a water dataset (see [Supplementary Table 1](#)), resulting in predictions that are in good agreement with experiment (see [Supplementary Figure 12](#)). Next, [Figure 3g–i](#) present a comparative analysis of calculated versus experimental lithium-ion self-diffusion coefficients for three thiophosphate solid-state electrolytes: $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, $\text{Li}_{10}\text{SiP}_2\text{S}_{12}$, and $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$, spanning temperatures from 300 to 1100 K. Notably, NEP89 achieves excellent agreement with AIMD simulations and a specialized force field [24], while outperforming other foundation models across the full temperature range.

Out-of-the-box large-scale MD simulations. After demonstrating the capabilities of NEP89 in dynamical simulations using small systems designed for benchmarking purposes, we next showcase the out-of-the-box applicability of NEP89 for large-scale atomistic MD simulations of inorganic and organic materials, which are typically impractical for other foundation models. The examples include a million-atom-scale compression simulation of the complex multicomponent $\text{Cu}_{0.7}\text{Mo}_{25.0}\text{Ta}_{29.6}\text{V}_{17.0}\text{W}_{27.7}$ (CuMoTaVW) alloy, calculations of ion diffusivities in water, a study of rocksalt dissolution kinetics, a simulation of methane combustion, and a dynamical simulation of a protein-ligand system (see Methods Sections ‘MD simulation of CuMoTaVW alloy’–‘MD simulation of protein dynamics’ for computational details).

For the CuMoTaVW alloy, common neighbor analysis ([Figure 4a](#)) shows that NEP89 successfully maintains structural stability during isothermal equilibration and compression. The polycrystalline body-centered cubic structure remains intact throughout the deformation process, with no observable phase transformation. The yield strength and yield strain calculated by NEP89 are 9.2 GPa and 5.8%, respectively ([Figure 4b](#)), which closely reproduce the experimentally measured yield strength (~ 10.0 GPa) and yield strain ($\sim 6\%$) [38]. This quantitative agreement illustrates the ability of NEP89 to accurately capture atomic interactions in complex multicomponent alloy systems.

Moreover, the diffusivities of various monovalent and divalent ions in water predicted by the NEP89 model ([Figure 4c](#)), and their corresponding activation energies ([Supplementary Figure 13](#)), are in good agreement with experiments [37, 39]. In combination with computational efficiency, this enables the prediction of, for example, the dissolution kinetics of a small salt crystallite in water ([Figure 4d,e](#), [Supplementary Figure 14](#)). The expected Arrhenius behavior of the dissolution process is captured out of the box with NEP89 ([Figure 4d](#), inset).

The methane combustion simulation at 3000 K ([Figure 4f](#)) shows that NEP89 generates primary reaction products and product distributions that are remarkably similar to those obtained with the ANI-1xn model [21], indicating the capability of NEP89 to capture the un-

derlying reaction physics and mechanisms. Atomic snapshots during combustion ([Figure 4g](#)) confirm that NEP89 also provides reasonable predictions of the reaction chemistry. Furthermore, NEP89 reproduces the near-exponential decay profiles of CH_4 and O_2 concentrations originally observed in ANI-1xn simulations [21], demonstrating its robust performance in modeling organic reactions. While the overall reaction pathways predicted by NEP89 and ANI-1xn [21] are consistent, NEP89 yields faster combustion kinetics. At 3000 K, the calculated reaction rate constant k is $2.90 \times 10^{15} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ for NEP89, approximately six times larger than the value obtained with ANI-1xn ($4.99 \times 10^{14} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$). As discussed in [Supplementary Section 12](#) and [Supplementary Figures 16–17](#), this kinetic acceleration is likely associated with a lower C–H bond dissociation energy for NEP89 relative to the ANI-1xn reference, leading to a reduced effective activation barrier for methane combustion.

For the protein-ligand case, we simulate the dynamics of a protein (T4 lysozyme L99A/M102Q, 3HTB) in complex with a ligand (2-propylphenol, JZ4) using NEP89 and compare the results with those obtained using the specialized protein-optimized CHARMM36 force field. As shown in [Figure 4h–j](#), both models achieve the stable binding of the ligand JZ4 to the protein 3HTB in long-term simulations. The binding energy calculated by NEP89 is -1.64 eV , while the reference value from CHARMM36 is -1.24 eV , suggesting that NEP89 can reasonably describe the protein-ligand interactions. In addition, we calculated the Ramachandran diagram of the protein obtained by the two potentials, which can be used to analyze whether the conformation of the protein model conforms to the rules of stereochemistry. As shown in the [Figure 4l,m](#), the total allowed areas calculated by NEP89 and CHARMM36 are 63.1% and 90.47%, respectively. While NEP89 shows a lower proportion of allowed areas compared to the specialized CHARMM36 force field, both Ramachandran diagrams [40] show that the highest probability of protein configuration is around the $(-60^\circ, -30^\circ)$ area, with a higher proportion of α area (bottom left area in [Figure 4l,m](#)) than that of β area (top left area in [Figure 4l,m](#)). These observations indicate that NEP89 captures protein conformational preferences at a qualitative level, but its ability to accurately reproduce the Gibbs free energy landscape of protein-ligand systems remains limited ([Supplementary Figure 19](#)). This limitation likely reflects incomplete coverage of protein-ligand configurations in the training data and/or the absence of an explicit treatment of long-range electrostatic interactions in the current NEP89 framework, highlighting a key direction for future development (see [Discussion](#) section). We also calculated the number of protein-water hydrogen bonds, obtaining 220 for NEP89 and 375 for CHARMM36, which are of the same order of magnitude. Overall, these results demonstrate that NEP89 can qualitatively capture certain aspects of protein-ligand interactions and protein structure, even

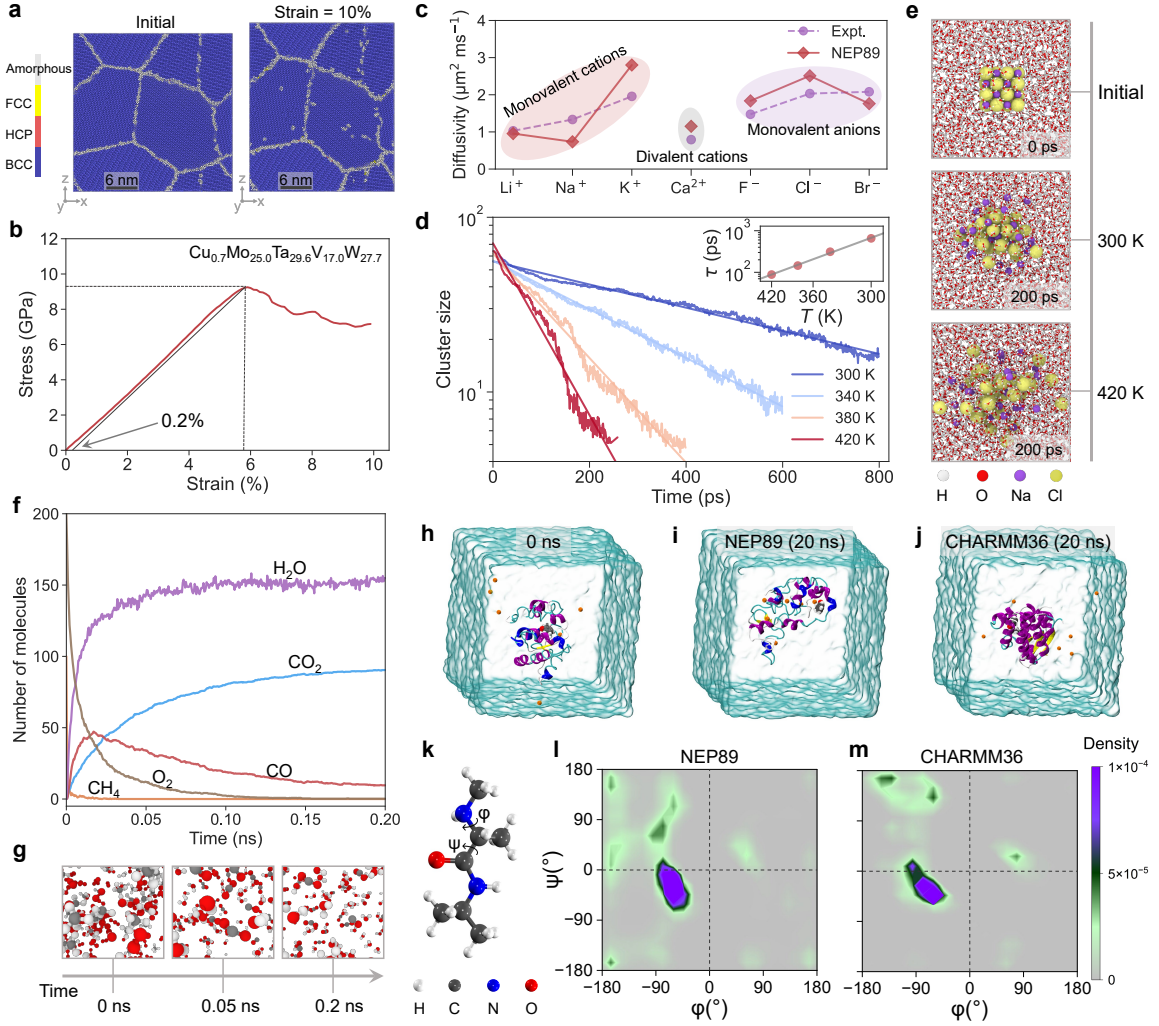


FIG. 4. **NEP89 applications in inorganic and organic systems.** **a**, Snapshots of atomic configurations of $\text{Cu}_{0.7}\text{Mo}_{25.0}\text{Ta}_{29.6}\text{V}_{17.0}\text{W}_{27.7}$ alloy at different stages during MD simulations (1 221 240 atoms) using NEP89. The left panel shows the initial configuration; the right panel shows the configuration under 10% strain, colored by common neighbor analysis. Dark blue, red, and yellow indicate atoms in body-centered cubic, hexagon close-packed, and face-centered cubic environments, respectively; off-white indicates atoms in amorphous environments. **b**, Stress-strain curve and yield strength predicted using NEP89. The dashed line indicates the 0.2% offset method used to determine the yield strength. **c**, Diffusivity of various ions in water at 300 K predicted by NEP89, compared to experimental values from Ref. [37]. **d**, Dissolution of NaCl nanocrystal in water at different temperatures calculated by NEP89. The dissolution rate τ is extracted from the power law fits (straight lines) at each temperature, and collected in the inset, illustrating that NEP89 captures the expected Arrhenius behavior of the dissolution process. **e**, Atomic snapshots of the spatial distribution of NaCl during dissolution in water at different temperatures. **f**, Time evolution of O_2 , CH_4 , and the major products (CO , CO_2 , H_2O) during a methane combustion simulation at 3000 K. **g**, Atomic snapshots of initial reactants, intermediate species, and final products. **h**, Initial protein-ligand model; **i,j**, Configurations obtained after 20 ns of MD simulations using NEP89 (**i**) and CHARMM36 (**j**). **k**, Schematic diagram of the backbone rotation angle (ϕ and ψ) used in the Ramachandran diagram. **l,m**, Ramachandran diagram showing the distribution of ϕ and ψ angles from simulations using NEP89 (**l**) and CHARMM36 (**m**).

though it is not specifically optimized for these protein systems. Further improvements are expected through fine-tuning, which we will introduce next.

Fine-tuning applications. For some specific applications, the accuracy of NEP89 may not be sufficient. In such cases, one can still fully leverage NEP89 to quickly obtain a reliable model through a simple but effective

active learning process. We can first use NEP89 to perform atomistic simulations at our target conditions, generating abundant trajectories. By sampling structures from the trajectories and performing single-point DFT calculations, we can create a training dataset to fine-tune NEP89 into a special-purpose model for the material and application in question. This process is considerably more efficient than sampling structures using AIMD.

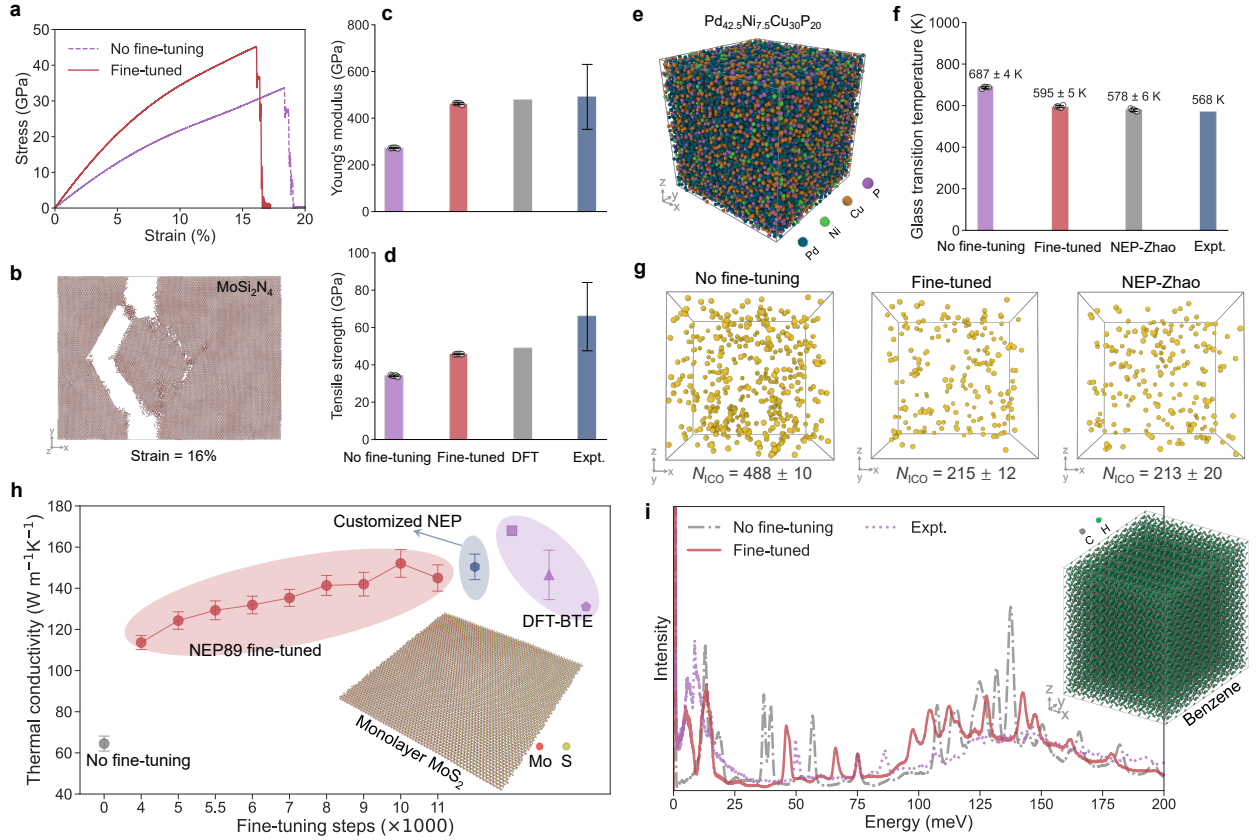


FIG. 5. **Example applications enabled by fine-tuning NEP89.** **a–d**, Mechanical properties of monolayer MoSi_2N_4 . **a**, Stress-strain curves averaged over five independent MD simulations at 300 K. **b**, Atomic snapshot at 16% strain after fine-tuning, showing fracture. **c,d**, Comparison of Young’s modulus and tensile strength before and after fine-tuning. Bar heights (here and in panel **f**) represent the mean of five independent simulations, error bars indicate the standard deviation (s.d.), and individual data points are shown as white circles with black outlines to indicate the data distribution. DFT and experimental data were taken from [41]. **e–g**, Thermal and structural properties of $\text{Pd}_{42.5}\text{Cu}_{30}\text{Ni}_{7.5}\text{P}_{20}$ metallic glass. **e**, Atomic snapshots of the quenched metallic glass structure generated by a fine-tuned model based on NEP89. **f**, Comparison of T_g values before and after fine-tuning, with results obtained using the NEP model by Zhao *et al.* [42] and experimental data [43] (no uncertainty reported). **g**, Spatial distribution and number (N_{ICO}) of icosahedral structures (colored in gold for central atoms) in a quenched glass predicted by different potentials. Numerical values in the panels denote the mean $\pm$ s.d. from five independent simulations. **h**, Thermal conductivity of monolayer MoS_2 . Results are compared with predictions from a customized NEP model [44] and Boltzmann transport equation (BTE) calculations using DFT-derived force constants [45–47]. Thermal conductivity values are presented as mean $\pm$ s.e.m. (standard error of the mean) across ten independent simulations. **i**, Simulated inelastic neutron scattering spectra for crystalline benzene at 127 K using NEP89 and a fine-tuned model, compared with experimental data [48].

In the following, we present four case studies, including the mechanical properties of the 2D material MoSi_2N_4 , thermal and structural properties of $\text{Pd}_{42.5}\text{Cu}_{30}\text{Ni}_{7.5}\text{P}_{20}$ metallic glass, thermal conductivity of monolayer MoS_2 , and simulated inelastic neutron scattering spectra for crystalline benzene (see Methods Sections ‘Fine-tuning for MoSi_2N_4 and MD simulations’–‘Fine-tuning for crystalline benzene and inelastic neutron scattering calculations’ for fine-tuning and simulation details).

MoSi_2N_4 has been successfully synthesized via chemical vapor deposition in recent years and exhibited promising properties [41]. Here, we showcase the fine-tuning of NEP89 specifically for MoSi_2N_4 with only 100 additional configurations. The out-of-the-box predictions by NEP89 for the fracture strength and the Young’s mod-

ulus are (272 ± 1) GPa and (34.2 ± 0.4) GPa, respectively. The fine-tuned model on the other hand yields (463 ± 4) GPa and (45.7 ± 0.2) GPa, respectively, in substantially better agreement with the experimental values of (491 ± 139) GPa and (66 ± 18) GPa, respectively (Figure 5a–d).

The $\text{Pd}_{42.5}\text{Ni}_{7.5}\text{Cu}_{30}\text{P}_{20}$ alloy is among the best-known bulk metallic glasses in terms of glass-forming ability [49]. Here, we demonstrate that fine-tuning NEP89 with only around 200 additional configurations yields excellent performance. After fine-tuning, the computed glass transition temperature T_g matches closely with both the prediction from a specialized NEP model [42] and experimental data [43], as shown in Figure 5e–f. Moreover, the icosahedral coordination, which describes the short-range

structure, is quantitatively improved after fine-tuning, achieving results comparable to a specialized NEP model (Figure 5g) [42].

Next, we demonstrate that fine-tuning the NEP89 model with only 104 additional configurations enables accurate prediction of the thermal conductivity of the monolayer MoS₂. Figure 5h shows the evolution of the predicted thermal conductivity using the homogeneous nonequilibrium MD method [50] as a function of the number of fine-tuning steps. As the latter increases, the predicted thermal conductivity gradually rises and eventually converges to the reference values. Notably, after 10 000 fine-tuning steps, the model yields thermal conductivity values that closely match those reported using a customized NEP potential [44] as well as the Boltzmann transport equation approach using force constants obtained via DFT calculations [45–47].

Finally, we showcase the prediction of scattering experiments using NEP89 and a fine-tuned version trained on 119 additional structures obtained via active learning. While the simulated spectrum from NEP89 is overall in qualitative agreement with experimental data [48] (Figure 5i), the peaks in the spectrum, which correspond to phonon modes, deviate from their experimental counterpart by about 10 meV. After fine-tuning, the predicted mode frequencies agree even better with the experimental spectrum, barring a general red shift of the broad feature at 100 meV, which can be attributed to the vdW-DF-cx exchange correlation functional used for the DFT calculations of the structures for fine-tuning. Moreover, the relative peak intensities are in better agreement with experiments as well.

To assess the robustness of NEP89 under varied fine-tuning tasks, we performed three independent runs with different random seeds across four representative systems (see Supplementary Figure 21). The resulting predictions remain highly consistent, indicating the stability and reliability of the fine-tuning scheme.

III. DISCUSSION

In summary, we have introduced NEP89, a highly efficient and accurate foundation model for large-scale atomistic simulations of both inorganic and organic materials across 89 elements. To the best of our knowledge, no existing machine-learned interatomic potential model offers this combination of elemental and chemical breadth with inference speeds comparable to typical empirical potentials. NEP89 represents a key milestone toward the long-standing goal of a unified computational framework that delivers near-ab-initio accuracy and near-empirical-potential computational efficiency across diverse material systems.

To achieve this, we employed an iterative training strategy incorporating descriptor-space subsampling, targeted data curation, and reference energy adjustments. The resulting training dataset strikes a balance

between diversity and compactness, supporting reliable predictions across a wide range of static and dynamic properties, and remains flexible for future extension.

Comprehensive benchmarks highlight the competitive performance of NEP89 in static properties, including formation energies, binding energies, adhesion energies, lattice energies, vacancy formation energies, phonon frequencies, and elastic moduli. NEP89 also demonstrates robust predictions in dynamic properties, such as bonding statistics in amorphous carbon, thermodynamic properties of liquid water, and lithium-ion transport in solid-state electrolytes. NEP89 often matches or outperforms representative foundation models, exhibiting good agreement with experimental results and robustness across diverse applications.

Beyond benchmarks on small-scale systems, we demonstrated the out-of-the-box capability of NEP89 in large-scale MD simulations, enabled by its computational efficiency, which is comparable to that of empirical potentials. From large-scale simulations of complex alloy systems and solution chemistry to modeling methane combustion and protein-ligand dynamics, NEP89 showcases versatility across a wide range of inorganic and organic systems. Notably, NEP89 achieves quantitative agreement with experiments on yield strength and ion diffusivities in water, reproduces the expected Arrhenius behavior in rocksalt dissolution kinetics, and matches specialized model predictions for combustion reaction dynamics. In addition, it qualitatively captures certain structural features and interactions in protein systems. These results highlight the ability of NEP89 to model complex systems with high fidelity, even without domain-specific optimization.

The fine-tuning capabilities of NEP89 provide a practical pathway for rapidly developing specialized models from limited training data. Researchers can efficiently customize NEP89 for specific materials or conditions, as demonstrated in case studies on MoSi₂N₄ mechanical properties, thermal and structural properties of Pd_{42.5}Cu₃₀Ni_{7.5}P₂₀ metallic glass, thermal conductivity of monolayer MoS₂, and neutron scattering simulations of crystalline benzene. Fine-tuning yields substantially improved agreement with results from quantum-mechanical calculations and experiments, demonstrating the utility of NEP89 as both a versatile out-of-the-box and an adaptable foundation model.

While NEP89 offers high computational efficiency and broad applicability, it is important to acknowledge its current limitations and clarify the scope in which its predictions are most reliable. The current NEP89 foundation model does not explicitly account for long-range interactions, which may affect quantitative accuracy in ionic, aqueous, and large biomolecular systems. Although NEP89 qualitatively captures protein conformational preferences, accurately reproducing detailed free energy landscapes for complex biomolecules remains challenging. This limitation likely reflects both the absence of an explicit treatment of long-range electrostatic

interactions and incomplete coverage of protein-ligand configurations in the currently available training data. Further efforts to construct more comprehensive training datasets will therefore be important, especially for organic and large-molecule systems. In addition, the current NEP89 foundation model does not include descriptors associated with variable atomic charges or magnetic degrees of freedom, which may restrict its current applicability to systems with substantial charge transfer, strong magnetism, or coupling to external electric or magnetic fields.

These limitations reflect the scope of the present NEP89 implementation rather than its fundamental constraints. Importantly, they identify clear directions for systematic extensions in future work. As a concrete example, we have recently developed and released a charge-extended NEP method [51] in the GPUMD package, which introduces environment-dependent atomic charges and explicit long-range electrostatic interactions, demonstrating that such learnable physics can be integrated in a consistent and scalable manner. Extensions to include magnetic degrees of freedom or other physics-informed descriptors are natural next steps. The architectural extensibility, combined with the near-empirical-potential efficiency of the NEP89 foundation model, provides a solid foundation for the practical development of more specialized and physics-informed models.

While the demonstrated out-of-the-box and fine-tuning applications are not exhaustive, they provide a solid foundation for future explorations in various application domains. Such efforts will further characterize the capabilities of NEP89 and identify potential areas for improvement. We expect that with ongoing extension and refinement, the out-of-the-box performance of NEP89 will continue to improve. Additionally, we anticipate the development of numerous domain-specific, medium-sized models fine-tuned from NEP89, enabling tailored solutions for a wide range of applications.

IV. METHODS

Benchmark calculations for static properties.

Static properties were calculated with the help of the GPUMD-WIZARD [52], ASE [53], CALORINE [54], PHONOPY [34], and MATCALC [55] packages. Detailed computational procedures can be found in [Supplementary Sections 5–10](#). Atomic snapshots were generated using OVITO [56].

Computational performance evaluation. We performed MD simulations with 1000 steps on a 20-element equiatomic alloy composed of Mn, Cr, Fe, Co, Ni, Cu, Ag, W, Mo, Nb, Al, Cd, Sn, Pb, Bi, Zn, Ge, Si, Sb, and Mg [57] to evaluate the computational efficiency of different foundation models. The initial structure of the alloy is face-centered cubic with randomly distributed atoms and a lattice constant of 3.7 Å. The Nosé-Hoover

chain thermostat [58] with a coupling time of 0.1 ps was used to control the temperature. Simulations were conducted on systems ranging from the smallest unit cell to progressively larger supercells until memory limitations were reached. Benchmarks were performed using GPUMD (version 4.0) [59] for NEP89, LAMMPS (version 28 Mar 2023) for MACE-MP-0, and LAMMPS (version 2 Aug 2023) [60] for both CHGNet and M3GNet.

MD simulation of amorphous carbon. A system with 512 atoms in a diamond structure with a given density undergoes an initial rapid melting process at 9000 K for 25 ps, followed by a relaxation at 5000 K for 5 ps. This is followed by a rapid quenching from 5000 K down to 1000 K in 0.4 ps, with further relaxation stages at 1000 K for 5 ps and subsequently at 300 K for 5 ps. The time step for integration is 0.5 fs. The Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.05 ps was used to control the temperature.

MD simulation of water. A system with 699 atoms with an initial density of 1.0 g cm^{-3} was used in the MD simulations of water. The system was equilibrated in the isothermal-isobaric ensemble at a given temperature and a pressure of 1 bar for 100 ps and then in the canonical ensemble for 50 ps, followed by a production run in the microcanonical ensemble for 25 ps. The time step for integration was 0.5 fs. The Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.1 ps and the stochastic cell rescaling barostat [62] with a coupling time of 1 ps and an elastic modulus of 2 GPa, were used to control the temperature and isotropic pressure, respectively. For each temperature, five independent runs were performed.

Lithium-ion diffusion in solid-state electrolytes.

MD simulations were performed on systems with 900 atoms for each of the three thiophosphate solid-state electrolytes: $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, $\text{Li}_{10}\text{SiP}_2\text{S}_{12}$, and $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$. Each system was equilibrated in the isothermal-isobaric ensemble at a given temperature and a pressure of 1 bar for 50 ps and then in the canonical ensemble for 100 ps, followed by a production run in the microcanonical ensemble for 200 ps. The time step for integration is 0.5 fs. The Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.1 ps and the stochastic cell rescaling barostat [62] with a coupling time of 1 ps and an elastic modulus of 100 GPa, were used to control the temperature and diagonal pressure, respectively. For each material and each temperature, three independent runs were performed.

MD simulation of CuMoTaVW alloy. The initial configuration comprises a body-centered cubic polycrystalline structure with 10 grains (average grain size about 12 nm), forming a $27 \text{ nm} \times 27 \text{ nm} \times 27 \text{ nm}$ simulation box containing 1 221 240 atoms. Using a hybrid MC/MD method, we performed 4 million Monte

Carlo attempts to obtain a low-energy configuration. The system was then equilibrated at 300 K for 100 ps under the isothermal-isobaric ensemble. Subsequently, uniaxial compression was applied along the x -axis at an engineering strain rate of $2 \times 10^8 \text{ s}^{-1}$, up to a maximum strain of 10%. The Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.1 ps and the stochastic cell rescaling barostat [62] with a coupling time of 1 ps and an elastic modulus of 100 GPa, were used to control the temperature and lateral pressure, respectively. Yield strength is determined using the commonly adopted 0.2% offset method, which identifies the point where the stress-strain curve intersects a line parallel to the linear (elastic) region, offset by 0.2% strain.

Ion diffusivity in water. To compute the diffusivity of Li^+ , Na^+ , K^+ , Ca^{2+} , F^- , Cl^- , and Br^- , 14 (7) monovalent (divalent) ions along with the corresponding number of OH^- or H^+ ions were randomly mixed with 14 000 H_2O molecules at an initial density of 1.0 g cm^{-3} . The systems were equilibrated in the isothermal-isobaric ensemble at a given temperature and a pressure of 0 bar for 200 ps using the Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.02 ps to control the temperature and the stochastic cell rescaling barostat [62] with a coupling time of 0.08 ps to control the isotropic pressure, followed by a production run in the microcanonical ensemble for 400 ps using a time step of 0.2 fs. During the latter run, the mean-square displacement was recorded, from which the diffusivity was calculated.

Dissolution kinetics of NaCl in water. A nanocrystal comprising 32 NaCl units was placed in a cell surrounded by 14 000 H_2O molecules at a density of 1.0 g cm^{-3} . The system was then evolved in the canonical ensemble at a given temperature for up to 2 ns using a time step of 0.2 fs. The Langevin thermostat [63] with a coupling time of 0.02 ps was used to control the temperature. We confirmed that the dissolution kinetics remain consistent when using a larger time step of 0.5 fs (see [Supplementary Figure 15](#)). The size of the largest cluster was tracked over time, where the clusters were defined by connectivity using a maximum Na-Cl bond cutoff of 4 Å calculated using OVITO [56]. 10 runs were carried out at each temperature, and the crystal size was averaged over these runs. The dissolution rate for each temperature was determined by fitting the time dependence of the cluster size to a simple exponential law.

MD simulation of methane combustion. CH_4 and O_2 molecules were mixed into a reactive system containing 100 CH_4 and 200 O_2 molecules, enclosed in a cubic cell of dimensions $37.5 \text{ Å} \times 37.5 \text{ Å} \times 37.5 \text{ Å}$, corresponding to a density of 0.25 g cm^{-3} . Each trajectory began with isothermal equilibration at 300 K; subsequently, the temperature was ramped to trigger combustion. The

integration was performed with a timestep of 0.1 fs to align with the settings used for the ANI-1xnr model [21]. As shown in [Supplementary Figure 18](#), the evolution of combustion products remains consistent when using a larger timestep of 0.5 fs. The Langevin thermostat [63] with a coupling time of 0.01 ps was used to control the temperature. The time-dependent combustion products are averaged over five independent MD trajectories.

MD simulation of protein dynamics. Dynamics of a protein (T4 lysozyme L99A/M102Q, 3HTB) in complex with a ligand (2-propylphenol, JZ4) was simulated using both GROMACS [64] with the CHARMM36 [65] (for 3HTB) and CGenFF [66] (for JZ4), as well as GPUMD with NEP89. The system consists of 33 917 atoms, and the simulation was performed for 20 ns in the isothermal-isobaric ensemble using a timestep of 0.5 fs. The last 15 ns were used for data analysis. The Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.05 ps and the stochastic cell rescaling barostat [62] with a coupling time of 0.5 ps and an elastic modulus of 2.2 GPa, were used to control the temperature and isotropic pressure, respectively.

Fine-tuning for MoSi_2N_4 and MD simulations. Using the NEP89 model, we conducted tensile loading MD simulations of MoSi_2N_4 using an orthorhombic cell with 210 atoms at 300 K and evenly sampled 100 structures from the trajectory. Then we performed single-point DFT-D3(BJ) calculations using the VASP package (version 6.5.1) [67, 68] with the projected augmented wave method and the generalized gradient approximation, along with the Perdew-Burke-Ernzerhof (PBE) functional [69]. A vacuum layer of 20 Å was used to model the two-dimensional systems. The plane-wave truncation energy was set to 500 eV, and a Γ -centered k -point grid with $2 \times 2 \times 1$ divisions was used. Using the new training dataset, we trained 5000 steps to obtain the fine-tuned model for MoSi_2N_4 . Using both the NEP89 and fine-tuned models, we performed tensile loading simulations for the monolayer MoSi_2N_4 , using an orthorhombic cell with 44 800 atoms. The engineering strain rate in the herringbone direction was $8.6 \times 10^7 \text{ s}^{-1}$. The thickness of the monolayer was taken as 10.7 Å in calculating the volume, following earlier literature [41]. The Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.1 ps and the stochastic cell rescaling barostat [62] with a coupling time of 1 ps and an elastic modulus of 500 GPa, were used to control the temperature and lateral pressure, respectively. We conducted five independent tensile loading simulations and averaged them to report the stress-strain curves and standard deviations.

Fine-tuning for PdCuNiP metallic glass and MD simulations. To perform fine-tuning for the $\text{Pd}_{42.5}\text{Cu}_{30}\text{Ni}_{7.5}\text{P}_{20}$ metallic glass, we carried out quenching simulations for three specific compositions:

$\text{Pd}_{40}\text{Cu}_{30}\text{Ni}_{10}\text{P}_{20}$, $\text{Pd}_{40}\text{Ni}_{40}\text{P}_{20}$, and $\text{Pd}_{42.5}\text{Cu}_{30}\text{Ni}_{7.5}\text{P}_{20}$ using the NEP89 model. The simulation protocol involved constructing a 108-atom supercell equilibrated at 1800 K in the isothermal-isobaric ensemble for 1 ns, followed by rapid quenching to 300 K at a cooling rate of $5 \times 10^{10} \text{ K s}^{-1}$. The structure was then equilibrated at 300 K for 1 ns. From the resulting trajectory, 209 configurations were sampled. Single-point calculations were performed using VASP package (version 6.5.1) [67, 68] with the PBE functional [69], a Γ -centered k -point grid with a spacing of $0.2 \text{ \AA}^{-1}$, a plane wave energy cutoff of 600 eV, and a threshold for the self-consistency loop of 10^{-6} eV . After labeling the 209 structures with DFT reference data, the NEP89 model was fine-tuned for 5000 steps. For the calculation of the glass transition temperature (T_g) and short-range order analysis of $\text{Pd}_{42.5}\text{Cu}_{30}\text{Ni}_{7.5}\text{P}_{20}$, a system containing 32 000 atoms randomly arranged in a cubic box with dimensions of $7.6 \text{ nm} \times 7.6 \text{ nm} \times 7.6 \text{ nm}$ was equilibrated in the isothermal-isobaric ensemble (zero pressure) at 1800 K for 1 ns with a timestep of 1 fs, followed by rapid cooling to 300 K at a rate of $5 \times 10^{10} \text{ K s}^{-1}$, and further equilibrated at 300 K for 1 ns (see [Supplementary Figure 20](#) for potential energy curves and T_g). The Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.1 ps and the stochastic cell rescaling barostat [62] with a coupling time of 1 ps and an elastic modulus of 100 GPa, were used to control the temperature and isotropic pressure, respectively. The polyhedral template matching method [70] with a root-mean-square deviation cutoff of 0.12 was employed to identify local structural motifs such as icosahedral clusters and other short-range order structures. The reported T_g and icosahedral cluster counts are averages of five independent simulations, with standard deviations included.

Fine-tuning for MoS_2 and thermal conductivity calculations. Using the NEP89 model, we performed a 6 ns isothermal-isobaric simulation of MoS_2 in an orthorhombic box containing 90 atoms at 300 K, and applied the farthest point sampling method to sample 104 configurations from the trajectory. Single-point DFT-D3(BJ) calculations were performed with the VASP package (version 6.5.1) [67, 68] with the PBE functional [69], a k -space with a spacing of $0.15 \text{ \AA}^{-1}$, a threshold for the self-consistency loop of 10^{-6} eV , and a plane wave energy cutoff of 520 eV. The dataset with 104 structures was used to fine-tune the NEP89 model, yielding models for monolayer MoS_2 with different fine-tuned steps. Using the homogeneous nonequilibrium MD method [50], we calculated the thermal conductivity of a MoS_2 model containing 11 484 atoms. The Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.1 ps and the stochastic cell rescaling barostat [62] with a coupling time of 1 ps and an elastic modulus of 20 GPa, were used to control the temperature and in-plane pressure, respectively. Ten independent 10 ns simulations were conducted, and the standard error was calculated from

the results.

Fine-tuning for crystalline benzene and inelastic neutron scattering calculations. Energies and forces for the active-learning training structures were evaluated using the vdW-DF-cx exchange-correlation functional [71]. Crystalline benzene was simulated using both the NEP89 and fine-tuned models in a 57 024-atom supercell at 127 K, with a timestep of 0.5 fs. First, the system was pre-equilibrated in the canonical ensemble for 5 ps using the Berendsen thermostat [72], with a temperature-coupling time of 0.05 ps. Next, a second pre-equilibration step in the isothermal-isobaric ensemble using the stochastic cell rescaling method [62] for 500 ps was performed, with a temperature-coupling time of 0.05 ps, a pressure-coupling time of 0.1 ps, an external pressure of 0 GPa, and elastic moduli of $C_{xx} = C_{yy} = C_{zz} = 40 \text{ GPa}$. Finally, a path-integral MD simulation with 32 beads for 50 ps performed with the same settings as the isothermal-isobaric equilibration step ensured accurate cell volumes at low temperature [73, 74]. The trajectory was written to file every 50 fs during the path-integral MD simulation, and the last 100 frames were subsequently extracted and averaged to yield the average, equilibrated simulation cell that was used for the production simulations. A short pre-production run of 5 ps in the canonical ensemble using the Bussi-Donadio-Parrinello thermostat [61] with a coupling time of 0.05 ps, followed by another pre-production simulation of 5 ps in the microcanonical ensemble was then performed to ensure the dynamics were equilibrated to the averaged simulation cell. Finally, a 1 ns microcanonical production run was performed, with positions saved every 3 fs to avoid aliasing artifacts when computing the dynamic structure factor. The dynamic structure factor $S(\mathbf{q}, \omega)$ was computed from the resulting trajectories following Ref. [48], briefly summarized here. $S(\mathbf{q}, \omega)$ was computed using DYNASOR [75, 76] for 2116 random $\mathbf{q}$ -points up to $|\mathbf{q}| = 14 \text{ \AA}^{-1}$. Results were smoothed with a Gaussian of width $0.01 \text{ \AA}^{-1}$ and averaged over spherical $|\mathbf{q}|$ shells to obtain $S(q, \omega)$. $S(q, \omega)$ was then weighted for neutron probes using species-specific scattering lengths and convolved with the kinematic constraint and resolution function of the TOSCA spectrometer at the ISIS Neutron and Muon Source, UK. Finally, the weighted $S(q, \omega)$ was integrated to yield the spectrum $S(\omega) = \int S(q, \omega) dq$. The TOSCA resolution function, implemented via EUPHONIC [77] and RESINS [78], is based on the ABINS module in MANTID [79].

Data availability: The NEP89 model has been made available as part of the GPUMD 5.0 distribution [59]. The associated training data and the source data for Figure 1 are available via Zenodo at <https://doi.org/10.5281/zenodo.19440423> (Ref. [80]). Source data for Figures 2–5 are provided with this paper.

Code availability: The source code for GPUMD (version 5.0) is available at the Zenodo repository <https://doi.org/10.5281/zenodo.18977569> [81] and the GitHub repository <https://github.com/brucefan1983/GPUMD/releases/tag/v5.0>.

Demonstration examples for both out-of-the-box applications and fine-tuning of the NEP89 model are available via Zenodo at <https://doi.org/10.5281/zenodo.19440423> [80]. A step-by-step fine-tuning tutorial is available at https://github.com/brucefan1983/GPUMD-Tutorials/tree/main/examples/26_fine_tune_NEP89.

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Author Contributions: T.L., K.X., S.C., and Z.F. conceived the original ideas and implementation. T.L. and K.X. created the visualizations for figures and performed comprehensive benchmarking. Z.F. and S.C. led training dataset curation, descriptor-space subsampling, iterative refinement, and model development, with T.L. contributing to parts of these efforts. Specific application cases were contributed by the following authors: E.L. for crystalline benzene; Z.C. and B.Z. for methane combustion and protein-ligand cases; R.Z. for multicomponent alloys; J.L. for energy benchmarking; E.B. for ion diffusivity and NaCl dissolution; B.T. for MoSi₂N₄; Y.W. for amorphous carbon; and K.S. for iron vacancy clusters. T.L. also participated in preparing and organizing all application cases. T.L., K.X., S.C., and Z.F. fully participated in and led the preparation and revision of the manuscript. All authors participated in the review and editing process and engaged in extensive discussions. Z.F., P.E., T.A-N., and J.X. managed funding acquisition, project administration, resources, and provided supervision throughout the project.

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